



Precision Boundary Of Relativistic Time Dilation In Low-Momentum Muon Decay Measurements

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Our Interest in Participating

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1. Abstract (~120 words)

Relativistic time dilation, a fundamental concept of Einstein's Special Theory of Relativity, states that moving muons have a longer lifetime in the lab frame. While this effect has been confirmed with high-momentum muons, the precision limits of time dilation measurements for low-momentum muons largely remains unexplored.

This project aims to measure the decay of low-momentum muons until measurements become uncertain due to instrumental limits. We will analyze the lifetime of low-momentum muons and compare measured lifetimes with relativistic predictions. Furthermore, we aim to discover the momentum threshold where time dilation measurements become unreliable and where the precision boundary for the instrument lies. This study will help understand both relativistic effects and detector precision in low-momentum particle measurements.

2. Scientific Background(~150 words)

Muons are unstable subatomic elementary particles, similar to electrons, but around 200 times heavier. They decay to three other particles after an average lifetime of $2.2\mu\text{s}$. Relativistic time dilation states that the lifetime of a moving particle appears longer from an outside perspective compared to its proper lifetime at rest.

For the highest precision of relativistic time dilation in muons lifetime, experiments are performed at high-momentum. This makes it easier to detect and measure accurately, producing clean and very precise measurements.

For Muons, the dilated lifetime is:

$$\tau = \gamma\tau_0 \text{ where } \gamma = 1/\sqrt{1 - v^2/c^2}, \text{ (eq.1)}$$

τ - dilated lifetime | τ_0 - proper lifetime | v - muon's velocity | c - speed of light.

While measurements of high-momentum muons are straightforward, experiments with low-momentum muons are difficult.

Measuring relativistic time dilation at low momentum is hard because the effect is smaller and requires high precision instruments to distinguish. Studying low-momentum muons can help us determine the precision limits of relativistic measurements and detector capabilities.

3. Research Problem and Objective (~ 150 words)

While relativistic time dilation for high-momentum muons has been measured accurately, low-momentum measurements still remain less precise. This is because, at low-momentum, the dilated lifetime is only slightly longer than the proper lifetime. This makes accurate measurements difficult due to detector resolution.

This experiment's objectives include:

1. Produce muons at several low-momenta.
2. Measure lifetime of low-momentum muons.
3. Compare and analyze measured lifetime with relativistic predictions.
4. Determine the momentum threshold at which time dilation becomes uncertain.
5. Determine detector precision boundary and limitations.

These objectives help determine the precision boundary of detectors and error margins compared to relativistic predictions.

4. Theoretical Framework (~250 words)

4.1 Experimental Design

This experiment plans to produce muons and measure their decay at several low momenta. It aims to investigate how systemic errors and detector resolution affects precision of relativistic time dilation and other measurements. The objective is to identify the precision boundary where the dilation effect becomes too small to measure and resolve properly.

Low-momentum muon beams are produced and enter the vacuum beam pipe at definite momentum.

Scintillator 1 (S1) gives the start signal by detecting incoming muons that entered the detector system. Scintillator 2 (S2) is an Additional detector that confirms muon's passage and helps reduce background noise (optional).

After passing through S1 and S2, the muon is stopped in a target material (aluminum or carbon) where the muon decays. Scintillator 3(S3) gives the stop signal by detecting electrons from the muon decay.

Time-To-Digital Converter measure time difference (Δt) between the start signal from S1 and stop signal from S3. The Data is passed to the Data Acquisition and Analysis it collects , plots and determines error margins and trends.

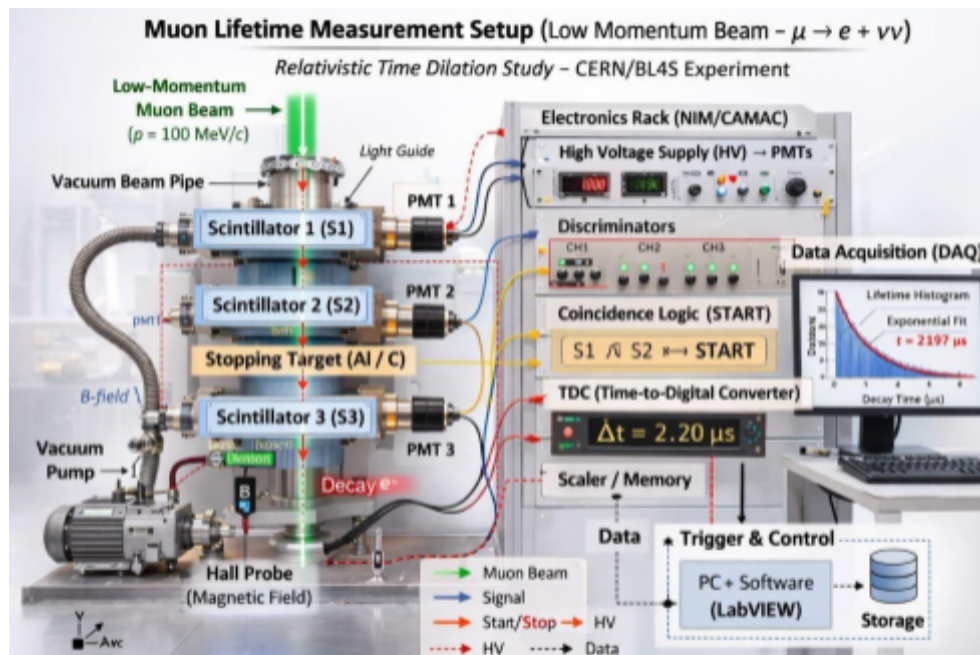


Figure 1: Muon Lifetime Measurement Setup
(FOR ILLUSTRATIVE PURPOSES ONLY)

4.2 Simulation Design

To complement the experimental setup, a simple simulation used to simulate muon decay and theoretical lifetime distributions. It was validated using a Monte Carlo simulation developed in python.

The code follows the following steps:

- It calculates the Relativistic Time Dilation (eq.1) for each momentum to determine the expected mean lifetime.
- For each momentum, 10,000 random decay events are generated using a logarithmic transformation (eq.2) to simulate realistic exponential decay.
- The simulated data is grouped into histogram and fitted with an exponential decay function (eq.3).
- The best-fit decay data is plotted into a curve.

The simulation precision is defined by the Standard Error of the Mean (SEM) (eq.4).

$$t = -\tau \ln(U) \text{ (eq.2)}$$

$$N(t) = N_0 e^{-t/\tau} \text{ (eq.3)}$$

$$SEM = \frac{\tau}{\sqrt{N}} = \frac{\tau}{\sqrt{10,000}} = 0.01 \times \tau \text{ (eq.4)}$$

U = Uniform Random Number | N = events (10,000) N_0 = Initial Number muons at $t = 0$ | τ - diluted lifetime

The simulation results are provided in Table 1 as follows:

Momenta (GeV/c)	Color	% Difference	Decay time	
			Theoretical	Simulated
1	Red	+2.51%	20.901 μ s	21.425 μ s
1.5	Yellow	+0.41%	31.255 μ s	31.382 μ s
2	Purple	+1.05%	41.628 μ s	42.067 μ s
2.5	Green	-1.30%	52.010 μ s	51.334 μ s
Table 1: Simulation Data				

The Python Code Used For The Simulation Is provided Below:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit

# Constants
tau0 = 2.197 #  $\mu$ s - normal muon decay time
m_mu = 0.1057 # GeV/c2 - muon mass

# Momenta to test (GeV/c) with corresponding colors
momenta = {1: "red", 1.5: "yellow", 2: "purple", 2.5: "green"}
N = 10000 # number of muons per event

# decay simulation according to exponential rule of muon decay
def exp_func(t, N0, tau):
    return N0 * np.exp(-t/tau)

# Main Loop
for p in momenta.keys():
    gamma = np.sqrt(1 + (p/m_mu)**2) # relativistic time dilation
    tau = gamma * tau0 # to find dilated decay time

    # Simulate decay times
    decay_times = -tau * np.log(np.random.rand(N)) #Artificially add randomness

    # make histogram
    hist, bins = np.histogram(decay_times, bins=50)
    bin_centers = (bins[:-1] + bins[1:])/2
    popt, _ = curve_fit(exp_func, bin_centers, hist, p0=[max(hist), tau])
    tau_fit = popt[1]

    print(f"Momentum: {p} GeV/c | Theory: {tau:.3f}  $\mu$ s | Simulated: {tau_fit:.3f}  $\mu$ s")
    # Plot on each event
    plt.plot(bin_centers, exp_func(bin_centers, *popt), 'r-', lw=2, c=momenta[p])

plt.xlabel("Decay time ( $\mu$ s)")
plt.ylabel("Counts")
plt.title(f"Muon Decay(GeV/c)")
plt.show() # show plotted graph
```

5. Data Analysis (~150 words)

To analyze the measurements of the experiment, decay-time data will be collected and represented using statistical plots. These plots help compare theoretical and measured data while also accounting for experimental uncertainties.

Figure 2 - Shows simulated muon decay curves for different momenta (table 1). Each curve is color graded according to its momentum and represents the distribution of muon decay. The steepest curve has the fastest decay time. That's because low momentum muons experience smaller time dilation making them decay faster.

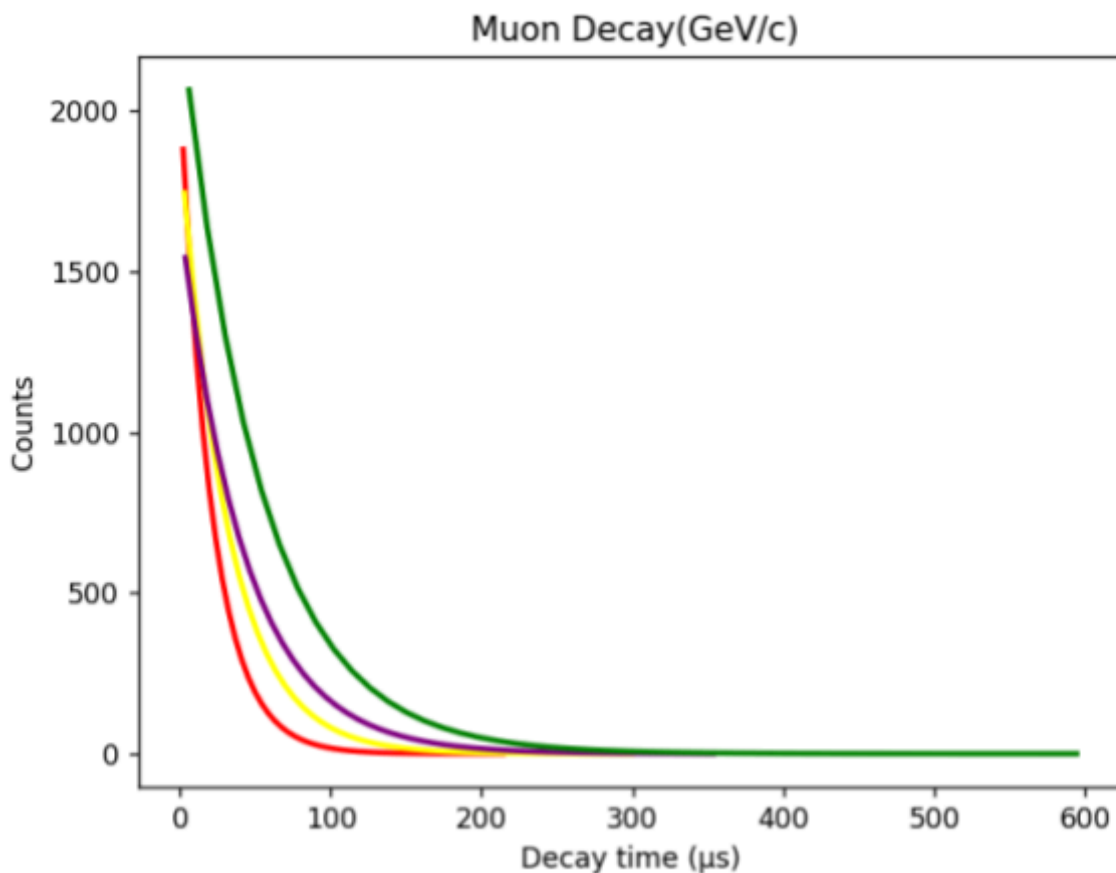


Figure 2: Muon Decay Curve For Different Momentum Muons

Simulation Plot from Table 1

6. Expected Result and Significance (~130 words)

This experiment plans to measure the lifetime of muons at several low momentum values. Based on relativistic time dilation, muons with higher momentum should decay more slowly and have a longer lifetime. On the contrary, muons with lower momentum should decay faster and have a shorter lifetime. As momentum gradually decreases, measured decay curves are expected to show gradual decrease in the effective muon lifetime.

However, when the momentum range becomes relatively low, the relativistic time dilation effect will be small. That is, at some point, the relativistic time dilation should become lower than our detector's resolution. As a result, measurements from our detector may deviate from theoretical curves. Therefore, this experiment aims to identify the momentum threshold and range at which measurements vary significantly from theoretical data rendering them scientifically inaccurate.

At the end of this experiment, the results will help determine the precision limit of the detector system used. Knowing these limitations is important to understand and improve particle detection techniques. Moreover, knowing when and why experimental data begins to deviate from theoretical predictions is an important aspect of evaluating science measurements.

7. Conclusion

This experiment proposes to measure the lifetime of muons at several low momenta in order to understand detector limits. By measuring muons with gradually decreasing momentum, the experiment plans to investigate how detector resolution affects the precision of measurements.

The results will help determine the precision boundary of the detection method. Understanding these limitations is important for scientifically accurate measurements. Furthermore, these limitations will help understand current detection flaws and improve particle detection methods.

Ultimately, this study will help understand relativistic physics and limits of reliable detection.

8. References

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